TEST PLAN

Pro Data – Control Panel

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INTRODUCTION

This document describes the approach and methodologies used by the testing team to plan, organize and perform the testing on Pro-Data web application.

The document can be used as a master test plan for preliminary testing and regression testing for each iteration of the sprint based on the decomposed testing parts provided below.

QA process in the project is performed over four major test items:

Test Items:

1. Pro-Data Control Panel - web application

SCOPE OF WORK

2.1 COMPONENTS AND FUNCTIONS TO BE TESTED

I. Test Item: **Pro-Data web application**

***Note: Component refers to Pro-Data Control Panel module - Function refers to each Pro-Data module sub-page

# ID	Component name	Function name	Test Suites link in QASE.io
1	Login	Login	Suite-3
2		Forgot Password	
3		Reset Password	
4	Services -> Cloud Platform	Header	
5		Cloud Platform Grid Table	Suite-1
6		Virtual Servers	
7		Load Balancer	Suite-16
8		Networks	Suite-19
9		Security Groups	
10		Disks	Suite-17
11		Backups	Suite-22
12		Custom Templates	Suite-18
13	Services -> Data Storage	Object Storage	
14	Services -> Containers	Managed Kubernetes	Suite-15
15	Account -> Profile	User Settings	Suite-23
16		User Management	Suite-24

17		Access Keys	
18	Account -> Project	Project Management	
19		Distribution of resources across projects	
20	Account -> Statistics	Actions Log	
21		Consumption	Suite-25
22	Account -> Security	Change Password	
23		2FA	
24	Account -> Notifications	Notification Settings	
25		Notification List	

***Note: Not all components and functions are fully covered with test cases - some of them might be in test design step or have none of it yet.

I. Test Item: **Pro-Data web application** – Test flows (i.e per user, location)

1. Functions/pages in which only one location should perform at
2. Functions/pages in which both locations should perform at
3. Functions/pages in which users from one project should perform at
4. Functions/pages in which users from multiple projects should perform at
5. Functions/pages in which root user should perform at
6. Functions/pages in which sub-user should perform at
7. Functions/pages in which only in English should perform at
8. Functions/pages in which only in Russian should perform at

2.2 COMPONENTS AND FUNCTIONS NOT TO BE TESTED

# ID	Component / Application / Device name	Function name	Reference
1	Server-side tasks	Virtual Servers	Virtual servers created for customers/test-users

RESOURCES

3.1. KEY PROJECT RESOURCES

#	Project Role	Name, e-mail
1	Директор по технологиям	Sergey Melikov sergey.melikov@pro-data.tech
2	Руководитель направления разработки ПО	Sergey Tixomirov t.sergey@pro-data.tech
3	Java программист	Sherzod Safarov sherzod@pro-data.tech
4	Java программист	Shoxrux Tashpulatov shoxrux@pro-data.tech
6	Frontend программист	Farxod Kodirov farxod@pro-data.tech
7	Frontend программист	Dmitriy Yushkin d.yushkin93@yandex.com
8	QA Engineer (Tester)	Jamshid Askarov, jamshid.a@pro-data.tech
9	Technical Support team	

3.2. TEST TEAM

#	Project Role	Name	Location	Responsibilities
1	QA (Tester)	Jamshid Askarov	Uzbekistan\ Tashkent	Test process creation, Test Result Reporting, Test Plan/Test cases/Checklist Creation, Bug Reporting, Bug verification, Scrum events Participation, Requirement Analysis, Automating test process

3.3. TEST ENVIRONMENT

#	Role	Resource	Hardware configuration	Software configuration
1	QA	Laptop (HP Probook)	16GB RAM, 512 SSD, AMD Ryzen 5 GHz	Windows 11, Google Chrome, MS Edge, Mozilla Firefox
2		PC	16GB RAM, i5 2.90 GHz	Windows 11, Google Chrome, MS Edge, Mozilla Firefox
3		iPhone 14 Pro Max	128GB, 100%	iOS v17.2.1
4		Galaxy Samsung tab	128GB	Android v12.1

3.4. TEST TOOLS

1	my.pro-data.tech	For performing post-release and regression testing
2	test-my.pro-data.tech	For performing pre-release and all other testing types
3	Wiki	For learning control panel features as KB
4	FAQ	For learning frequently asked questions
5	Bitrix	For bug reporting and QA-tasks
6	Qase	For test case creation
7	Confluence	For tracking release notes, CR documents
8	Yandex mail	For obtaining and installing apks
9	Telegram -> Разработка Pro-Data group	For communication with dev team
10	Telegram -> Pro-Data Customer Panel group	For communication about product with technical support team
11	Telegram -> QA. Автоматизация тестирования group	For communication about autotests

TEST DOCUMENTATION AND DELIVERABLES

#	Title	Responsible role(s)	Frequency (delivery time)	Method of delivery
1	Pro-Data Test Plan	QA Engineer (Tester)	Once, before the testing starts	Bitrix
2	Bug Report		Daily, during the sprint	Bitrix
3	Test Cases		Regularly, during the sprint	QASE.io

TEST STRATEGY

The testing of the Pro-Data web application will follow both "black box" and “white box” techniques as most of the checks have been automated and can give corresponding testing results regularly after each release/build; though some of the high-level MAT and AT level checks as well as test scenarios are performed manually that relies solely on the requirements / specifications and test cases.

The testing of the Pro-Data Control Panel application not only includes testing functionalities of the web, but also will follow verifying Cloud system, Virtual Servers and other parts of related Cloud Computing features manually on all levels: component, system, integration, and acceptance.

Both hardware and software-related checks of the device may be performed.

The testing of the Pro-Data web application should be performed at all test levels.

4.1 ENTRY CRITERIA

In the event of any of the following occurrences, the Testing Team reserves the right to halt partial or full-testing activities on a given release that usually occurs during release week:

- It is impossible to successfully run the web application following the instructions.
- The application lacks the specified change(s).
- The newly claimed functionality does not work or works improperly.
- A severe problem has occurred that does not allow testing to continue.
- Development has not corrected the problem(s) that previously suspended testing.
- Auto tests started failing in massive numbers

4.2 TEST METHODS

Testing is the process of attempting to find discrepancies between the program and its functional specification / requirements. The goal is to make sure that all functions of the Pro-Data Control Panel web application work correctly.

- Manual functional testing – is considered as the main method of application testing.
- Automation testing – is considered as partial method of application testing (supports only automated test cases)

4.3 TEST TYPES

Test Type Name	Short Description
Functional testing	Testing that verifies that the application functionality works correctly (the correct implementation of functional requirements)
Manual testing	Testing, where test cases are performed manually by a human being without the use of automation
Automation testing	Testing, where test cases are performed automatically by a test tool without the use of human interaction
Test cases-based testing	Testing is performed based on previously prepared test cases, test suites, and other documentation
Static testing	Testing without code execution
NFT testing	Testing new features of the application
Positive testing	Testing the application in a situation where all activities are carried out strictly as instructed, with no errors, deviations, incorrect data input, etc.
Negative testing	Testing the application in situations when whether operations (sometimes incorrect ones) performed, or data used may potentially lead to errors
Regression testing	Testing aimed at verifying the fact that previously working functionality has not been affected by errors caused by changes in the application or its environment
Re-testing (Verification testing)	Execution of test cases that have previously detected defects to confirm that the defect has been resolved
Module testing	Process that involves testing each unit of the modules or components to ensure they conform to the highest coding standards.
Integration testing	Testing the interaction between several parts of an application, i.e Web and Servers
Usability testing	Assessing whether the end-user can effectively utilize the product and their overall satisfaction with using the product.
GUI testing	Testing the software or system user interface by checking screens and controls like menus, buttons, icons, alignment, etc.
Performance testing	Involves activities about how well app is performed for users regarding without any delay and hangings while active use of it
Accessibility testing	Testing the application works for disabled people as well for common functions
Localization testing	Verify that the product has been adapted correctly and with high quality for usage in a specific language, i.e Ru and En
Compatibility testing	Testing the application across the different devices available required by the customer
Cross-browser testing	Testing the application across the browsers required by most of the customers (Chrome, Mozilla Firefox)
Cross-platform testing	Testing the application on different OS versions (Android, iOS)

Installability testing	Testing that app successfully creates cloud-related elements such as servers, machines and they are launched as expected